Light

# GCSE Computer Science – Network Protocols (Set 1)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A smartphone made by one company successfully exchanges data with a laptop made by a different company. What enables this communication between devices from different manufacturers?

- ○ **A.** Network protocols that provide standardised rules both devices follow
- ○ **B.** Special translation hardware installed in both devices
- ○ **C.** The devices must secretly use identical internal components
- ○ **D.** A physical cable that converts signals between brands

Q2 (1 mark)

Why are network protocols necessary for communication between computers?

- ○ **A.** They make computers run faster
- ○ **B.** They increase the storage capacity of computers
- ○ **C.** They provide agreed rules so different devices can understand each other
- ○ **D.** They reduce the cost of network cables

Q3 (1 mark)

When a user types 'www.example.com' into their browser and presses enter, which protocol requests the webpage from the web server?

- ○ **A.** HTTP or HTTPS requests the webpage from the server
- ○ **B.** SMTP requests the webpage from the server
- ○ **C.** IMAP requests the webpage from the server
- ○ **D.** TCP requests the webpage directly

Q4 (1 mark)

A shopping website displays a padlock icon in the browser's address bar. What does this indicate about the protocol being used?

- ○ **A.** The website is using HTTP which is always secure
- ○ **B.** The website cannot accept credit card payments
- ○ **C.** The website is blocking all cookies
- ○ **D.** The website is using HTTPS which encrypts data during transmission

Q5 (1 mark)

What is the main characteristic that makes TCP a 'reliable' protocol?

- ○ **A.** It transmits data faster than any other protocol
- ○ **B.** It automatically encrypts all data
- ○ **C.** It only works with wired connections
- ○ **D.** It ensures all data arrives complete and in the correct order

Q6 (1 mark)

What are the two main functions of the Internet Protocol (IP)?

- ○ **A.** Addressing packets and routing them across networks
- ○ **B.** Encrypting data and blocking viruses
- ○ **C.** Displaying web pages and playing videos
- ○ **D.** Sending emails and managing contacts

Q7 (1 mark)

When you click 'Send' on an email, which protocol transmits the message to the email server?

- ○ **A.** HTTP
- ○ **B.** IMAP
- ○ **C.** IP

- ○ **D.** SMTP

Q8 (1 mark)

What is the main advantage of using IMAP for accessing emails?

- ○ **A.** Emails remain on the server and can be accessed from multiple devices
- ○ **B.** Emails are automatically deleted after reading
- ○ **C.** Emails are sent faster than with other protocols
- ○ **D.** Emails are automatically encrypted

Q9 (1 mark)

At which layer of the TCP/IP model do HTTP, HTTPS, SMTP, and IMAP operate?

- ○ **A.** Link layer
- ○ **B.** Internet layer
- ○ **C.** Transport layer
- ○ **D.** Application layer

Q10 (1 mark)

TCP manages reliable delivery by sequencing packets and requesting retransmission of lost data. At which TCP/IP layer does this occur?

- ○ **A.** Application layer
- ○ **B.** Transport layer
- ○ **C.** Internet layer
- ○ **D.** Link layer

Q11 (1 mark)

IP assigns addresses to packets and determines routes across networks. At which TCP/IP layer do these addressing and routing functions occur?

- ○ **A.** Application layer
- ○ **B.** Transport layer
- ○ **C.** Link layer
- ○ **D.** Internet layer

Q12 (1 mark)

What is found at the link layer of the TCP/IP model?

- ○ **A.** Web browsers and email applications
- ○ **B.** IP addresses and routing tables
- ○ **C.** Network hardware such as NICs and device drivers
- ○ **D.** Encryption and authentication protocols

Q13 (2 marks)

A travel agency's booking system experiences network congestion, causing some data packets to be lost during transmission. How does TCP handle this situation?

- ○ **A.** TCP ignores the lost packets and delivers incomplete data
- ○ **B.** TCP cancels the entire transmission and displays an error
- ○ **C.** TCP detects missing packets through sequence numbers and requests their retransmission until all data arrives
- ○ **D.** TCP automatically switches to a wireless connection

Q14 (2 marks)

A podcast streaming service sends audio data from servers in Ireland to a listener in Japan. Describe the role of IP in this transmission.

- ○ **A.** IP adds source and destination addresses to packets and determines the route they take across international networks
- ○ **B.** IP compresses the audio files to reduce their size
- ○ **C.** IP encrypts the audio data to prevent piracy
- ○ **D.** IP converts the audio into a format the listener's device can play

Q15 (2 marks)

A newspaper website is considering switching from HTTP to HTTPS. What security benefit would this provide for readers?

- ○ **A.** Data transmitted between readers' browsers and the website would be encrypted, protecting it from interception

- ○ **B.** Readers would be able to access the website faster
- ○ **C.** The website would automatically block all advertisements
- ○ **D.** Readers would no longer need to create accounts to access content

Q16 (2 marks)

An insurance company employee sends a claim confirmation email from their desktop computer and later checks for a customer reply on their tablet. Explain how SMTP and IMAP are used.

- ○ **A.** SMTP retrieves the original email; IMAP sends the reply
- ○ **B.** SMTP and IMAP are both used to send emails simultaneously
- ○ **C.** SMTP sends the outgoing confirmation email; IMAP retrieves the reply from the server to the tablet
- ○ **D.** IMAP handles all email functions; SMTP is not used

Q17 (2 marks)

An online auction website allows users to bid on items using their web browsers. At which TCP/IP layer does this interaction occur, and why?

- ○ **A.** Link layer, because physical cables transmit the bid data
- ○ **B.** Internet layer, because IP addresses identify the users
- ○ **C.** Transport layer, because TCP ensures bids are transmitted reliably

○ **D.** Application layer, because web browsers and the auction website are user-facing network applications

Q18 (2 marks)

What are the main functions performed at the transport layer of the TCP/IP model?

○ **A.** Displaying web pages and running email programs

○ **B.** Converting data into electrical or optical signals for transmission

○ **C.** Setting up communication between hosts and managing reliable data delivery through packet sequencing

○ **D.** Assigning IP addresses and routing packets across networks

Q19 (2 marks)

A weather forecasting service sends data from sensors across a country to a central processing facility. What functions does the internet layer perform in this transmission?

○ **A.** It displays the weather data on forecasters' screens

○ **B.** It physically transmits signals through cables to each sensor

○ **C.** It addresses data packets and routes them from sensors across multiple networks to the central facility

○ **D.** It compresses the weather data to reduce file sizes

Q20 (2 marks)

What role does the network interface card (NIC) play at the link layer of the TCP/IP model?

- ○ **A.** It runs web browsers and email applications
- ○ **B.** It assigns IP addresses to devices on the network
- ○ **C.** It provides the physical connection to the network and transmits data as electrical, optical, or radio signals
- ○ **D.** It breaks large files into smaller packets for transmission

Q21 (2 marks)

A photography studio uploads high-resolution images to a client's cloud storage. How do TCP and IP work together in this process?

- ○ **A.** TCP compresses the images while IP encrypts them
- ○ **B.** TCP and IP both handle addressing—they are interchangeable
- ○ **C.** IP addresses and routes packets across networks; TCP ensures all image data arrives complete through reliable delivery
- ○ **D.** IP uploads the images while TCP downloads them

Q22 (2 marks)

Online banking websites always use HTTPS rather than HTTP. Explain why this is essential for banking transactions.

- ○ **A.** HTTPS makes banking websites load faster than HTTP
- ○ **B.** HTTPS allows banks to display their logo in the browser
- ○ **C.** HTTPS encrypts sensitive data like account numbers and passwords, preventing interception during transmission

○ **D.** HTTPS is required by law for all websites, not just banking

Q23 (2 marks)

A publishing company uses computers from three different manufacturers, running two different operating systems, yet they can all communicate over the company network. What makes this possible?

○ **A.** All computers must secretly be identical inside despite appearing different

○ **B.** The company purchased a special translation device for each computer

○ **C.** Network protocols provide standardised rules that all devices follow regardless of manufacturer or operating system

○ **D.** Only computers from the same manufacturer can actually communicate

Q24 (3 marks)

A scientific research facility sends experimental data from a laboratory computer to a colleague at a university overseas. Describe the order in which data passes through the four TCP/IP layers when being sent.

○ **A.** Link → Internet → Transport → Application: data starts at hardware and moves up to applications

○ **B.** Application → Transport → Internet → Link: research software prepares data, TCP segments it, IP addresses and routes it, the NIC transmits physical signals

○ **C.** Transport → Internet → Application → Link: TCP processes data before applications are involved

○ **D.** Internet → Link → Transport → Application: IP addressing

happens first before any other processing

Q25 (3 marks)

When a user's device receives streaming video from an entertainment service, data passes through the TCP/IP layers in a specific order. Describe this process from initial arrival to final display.

- ○ **A.** Application → Transport → Internet → Link: the video app sends a request first
- ○ **B.** Link → Internet → Transport → Application: physical signals arrive at the NIC, IP checks addressing, TCP reassembles packets, the video application displays the content
- ○ **C.** Transport → Application → Link → Internet: TCP handles data before it reaches any other layer
- ○ **D.** Internet → Transport → Link → Application: routing happens before physical signals are received

Q26 (3 marks)

An art gallery sends an email with attached exhibition images to a journalist, who reads it on three different devices. Explain the complete role of both SMTP and IMAP in this scenario.

- ○ **A.** SMTP handles everything on the gallery's side; IMAP handles everything on the journalist's side
- ○ **B.** SMTP transmits the email and attachments from the gallery to email servers; IMAP allows the journalist to retrieve and view the same email from multiple devices because it keeps messages stored on the server
- ○ **C.** SMTP and IMAP both transmit the email simultaneously to ensure faster delivery

○ **D.** IMAP sends the original email; SMTP is only used for replies

Q27 (3 marks)

A government portal allows citizens to submit tax returns online. These submissions must arrive complete and accurate. Explain how TCP's reliability mechanisms ensure this.

○ **A.** TCP encrypts tax data so it cannot be read by unauthorised parties

○ **B.** TCP compresses tax returns to make them transmit faster

○ **C.** TCP assigns sequence numbers to packets, sends acknowledgments for received data, and automatically retransmits any lost packets until the complete tax return is confirmed received

○ **D.** TCP sends each tax return through multiple different routes simultaneously

Q28 (3 marks)

A freelance marketplace handles profiles (public information), messaging (private conversations), and payments (financial transactions). Evaluate when HTTP versus HTTPS should be used for different parts of this website.

○ **A.** HTTP for everything—HTTPS is only needed for government websites

○ **B.** HTTPS for everything—all modern websites should encrypt all traffic regardless of content sensitivity

○ **C.** HTTP for profiles and messaging, HTTPS only for payments

○ **D.** HTTPS for payments only, HTTP for profiles, no protocol needed for messaging

Q29 (3 marks)

An international shipping company tracks packages using data sent between offices in Singapore, Germany, and Brazil. Explain how IP enables this communication across such geographically dispersed locations.

○ **A.** IP requires direct cable connections between all three countries

○ **B.** IP assigns unique addresses to each office's systems and routes packets across multiple interconnected networks, finding paths through various intermediate points regardless of physical distance

○ **C.** IP can only work within a single country; separate protocols handle international communication

○ **D.** IP sends identical copies of all data to every country simultaneously

Q30 (3 marks)

When a user accesses their online medical portal to view test results, multiple protocols work together across TCP/IP layers. Explain how HTTP/HTTPS, TCP, and IP each contribute to delivering the test results to the user's screen.

○ **A.** They all perform the same function but at different speeds

○ **B.** Only HTTPS is needed; TCP and IP are outdated protocols

○ **C.** HTTPS at the application layer handles the secure web page request, TCP at the transport layer ensures reliable delivery of all data, IP at the internet layer addresses and routes packets across networks to reach the user

○ **D.** IP encrypts the medical data, TCP compresses it, HTTP displays it on screen

Total: **55** marks • Time: **30 min**

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